

STEP 1: Prepare Your data

Lab ID,	Age (BP),	Error (±),	Depth (cm)
UBA-1234,	1200,	30,	10
UBA-1235,	2500,	45,	25
UBA-1236,	3800,	50,	40
UBA-1237,	5100,	60,	55
UBA-1238,	6400,	70,	70

STEP 2: Run Rbacon

```
# Basic run with default settings
Bacon("YourCore")
```

```
# Run with custom settings (recommended) Bacon("YourCore",
```

```
acc.mean = 20, # prior mean accumulation rate (yr/cm) acc.shape = 1.5, # shape of gamma distribution for acc rate
mem.mean = 0.7, # memory mean (how much adjacent sections correlate)
mem.strength = 4, # strength of memory prior
thick = 5, # thickness of each depth section (cm)
d.min = 0, # minimum depth
d.max = 80, # maximum depth
BCAD = FALSE) # use BP timescale (not BC/AD)
```

STEP 3: Extract Age Estimates

```
# Get age estimate at a specific depth
Bacon.Age.d(20) # median age at 20 cm depth
```

```
# Get full summary at specific depths depths <- seq(0, 80, by = 5) # every 5 cm
```

```
# Extract age with confidence intervals
age_summary <- sapply(depths, function(d) {
ages <- Bacon.Age.d(d)
c(depth = d,
mean = mean(ages),
median= median(ages),
lo95 = quantile(ages, 0.025),
hi95 = quantile(ages, 0.975))
})
```

```
# Convert to dataframe age_df <- as.data.frame(t(age_summary)) print(age_df)
```

STEP 4: Extract Accumulation Rate (Rbacon's Native Output)

```
# Accumulation rate at a specific depth (years/cm)
acc <- accrate.depth(d = 20)

# Summary statistics
mean(acc)           # mean accumulation rate
median(acc)         # median
quantile(acc, c(0.025, 0.975)) # 95% confidence interval

# Accumulation rate across all depths
depths <- seq(0, 80, by = 5)

acc_summary <- sapply(depths, function(d) {
  a <- accrate.depth(d)
  c(depth = d,
    mean  = mean(a),
    median = median(a),
    lo95  = quantile(a, 0.025),
    hi95  = quantile(a, 0.975))
})

acc_df <- as.data.frame(t(acc_summary))
print(acc_df)
```

STEP 5: Calculate Sedimentation Rate Properly (Key Step)

```
# -----
# WRONG WAY — loses uncertainty
# -----
sed_rate_wrong <- 1 / mean(accrate.depth(d = 20))

# -----
# CORRECT WAY — invert all MCMC iterations
# -----

# Step 8a — Extract full MCMC distribution at each depth
depths <- seq(0, 80, by = 5)

sed_rate_summary <- sapply(depths, function(d) {

  # Get full MCMC distribution of accumulation rate
  acc_iterations <- accrate.depth(d)

  # Invert ALL iterations (not just the mean!)
  sed_iterations <- 1 / acc_iterations

  # Summarize properly
  c(depth = d,
    mean  = mean(sed_iterations),
    median = median(sed_iterations),
    lo95  = quantile(sed_iterations, 0.025),
    hi95  = quantile(sed_iterations, 0.975))
})

# Convert to dataframe
sed_df <- as.data.frame(t(sed_rate_summary))
colnames(sed_df) <- c("Depth_cm", "Mean_SedRate",
  "Median_SedRate", "Lo95", "Hi95")
print(sed_df)
```

STEP 6: Plot Sedimentation Rate With Uncertainty

```
# Plot sedimentation rate with 95% confidence envelope
plot(sed_df$Depth_cm, sed_df$Mean_SedRate,
     type = "l",
     col = "blue",
     lwd = 2,
     xlab = "Depth (cm)",
     ylab = "Sedimentation Rate (cm/yr)",
     main = "Sedimentation Rate with 95% CI",
     ylim = range(sed_df$Lo95, sed_df$Hi95))
```

```
# Add 95% confidence envelope
polygon(c(sed_df$Depth_cm, rev(sed_df$Depth_cm)),
       c(sed_df$Hi95, rev(sed_df$Lo95)),
       col = rgb(0, 0, 1, 0.2),
       border = NA)
```

```
# Add median line
lines(sed_df$Depth_cm, sed_df$Median_SedRate,
     col = "darkblue", lty = 2)
```

```
# Add legend
legend("topright",
     legend = c("Mean", "Median", "95% CI"),
     col = c("blue", "darkblue", rgb(0,0,1,0.2)),
     lty = c(1, 2, NA),
     fill = c(NA, NA, rgb(0,0,1,0.2)))
```

Always use (1 / all_iterations) not (1 / mean) for sedimentation rate — this preserves the full uncertainty from Rbacon's Bayesian analysis.

Careful Points

```
library(rbacon)

# Run Bacon
Bacon("YourCore",
      acc.mean = 20,
      mem.mean = 0.7,
      thick    = 5,
      d.min    = 222,
      d.max    = 432,
      BCAD     = FALSE)

# Extract outputs for ML training
depths <- seq(222, 432, by = 1) # every 1 cm

results <- sapply(depths, function(d) {
  ages <- Bacon.Age.d(d)
  acc  <- accrate.depth(d)
  sed  <- 1 / acc

  c(depth      = d,
      mean_age  = mean(ages),
      median_age = median(ages),
      lo95      = quantile(ages, 0.025),
      hi95      = quantile(ages, 0.975),
      ci_width  = quantile(ages, 0.975) - quantile(ages, 0.025),
      acc_rate  = mean(acc),
      sed_rate  = mean(sed))
})

# Convert to dataframe
ml_training_data <- as.data.frame(t(results))

# Export for ML
write.csv(ml_training_data, "ML_training_data.csv", row.names = FALSE)
```

```
library(rbacon)

Bacon("CoreB",
      acc.mean = 50,      # higher mean — wider depth span, older ages
      mem.mean = 0.7,
      thick    = 10,      # thicker sections — fewer dates over longer core
      d.min    = 175,
      d.max    = 909,
      BCAD     = FALSE)

Bacon.Age.d()
→ built into rbacon package
→ works only after Bacon() is run
→ reads MCMC iterations from memory

accrate.depth()
→ built into rbacon package
→ works only after Bacon() is run
→ reads MCMC iterations from memory

Alternative
→ read CoreA_ages.txt file directly
→ works anytime
→ only gives mean/median/CI (no full distribution)
```

MLP Setup

INPUT (X) — What ML Learns From

From your **lab data** (what you already have):

- 1. Depth (cm) ← most important input
- 2. age (BP) ← at dated depths only
- 3. Error ($\pm$ yr) ← measurement uncertainty

That is it — because that is **all you are feeding into Rbacon**

OUTPUT (Y) — What ML Predicts

From your **Rbacon run**:

- 1. Mean age (cal BP)
- 2. Median age (cal BP)
- 3. Lower 95% CI (cal BP)
- 4. Upper 95% CI (cal BP)
- 5. CI width (uncertainty)
- 6. Accumulation rate (yr/cm)
- 7. Sedimentation rate (cm/yr)

What Your Training Data Should Looks Like

Each row = one depth (every 1 cm from Rbacon output)

depth | age | error | mean_age | median_age | lo95 | hi95 | ci_width | acc_rate | sed_rate

